Virtual Networking

Private IP Address (Local Address – LA)

Definition:
A **Private IP address** is used **inside a local network** and is **not visible on the Internet**.

Private IP Address – Roll Numbers inside the College

Inside the college:

- Every student has a **roll number**
- Roll numbers are meaningful **only inside the college**
  Student A → Roll No: 21
  Student B → Roll No: 22

These roll numbers are like **Private IP addresses**.

Public Ip address

Definition:
A **Public IP address** uniquely identifies a network or device on the **Internet**.

Eg: The entire college has **one official postal address**:

TKM college of Engineering
Kollam – 6000XX

- Outside people do NOT know individual roll numbers
- They only know the **college address**

This postal address is the **Public IP address**.

3. Virtual LAN (VLAN)

VLAN in Virtualization?

In **virtualization**, a **VLAN** is used to **logically separate virtual machines (VMs)** and physical systems **over the same physical network infrastructure**.

Even though multiple VMs share the **same physical NIC, switch, and cables**, VLANs keep their **network traffic isolated**.

In a virtualized environment (like VMware vSphere, Microsoft Hyper-V, or KVM), the "cables" are virtual, and the "switch" is software.

The Concept: The Virtual Switch (vSwitch)

In virtualization, the physical server (Host) runs a software layer called a **Hypervisor**. Inside this Hypervisor resides a **Virtual Switch (vSwitch)**.

- **Physical World:** You configure VLANs on the physical switch ports.
- **Virtual World:** You configure VLANs on **Port Groups** inside the vSwitch.
- **Function:** The vSwitch acts as a bridge between the Virtual Machines (VMs) and the physical network adapter (NIC) of the server.

A MAC address is a permanent physical address of a device, whereas an IP address is a logical address assigned by the network. Private IPs are used inside local networks, while public IPs are used to identify networks on the Internet.

In a computer lab, the MAC address uniquely identifies each computer hardware, the logical (private) IP address identifies the system within the lab network, and the public IP address represents the lab on the Internet.

Mapping of Logical address to Physical address

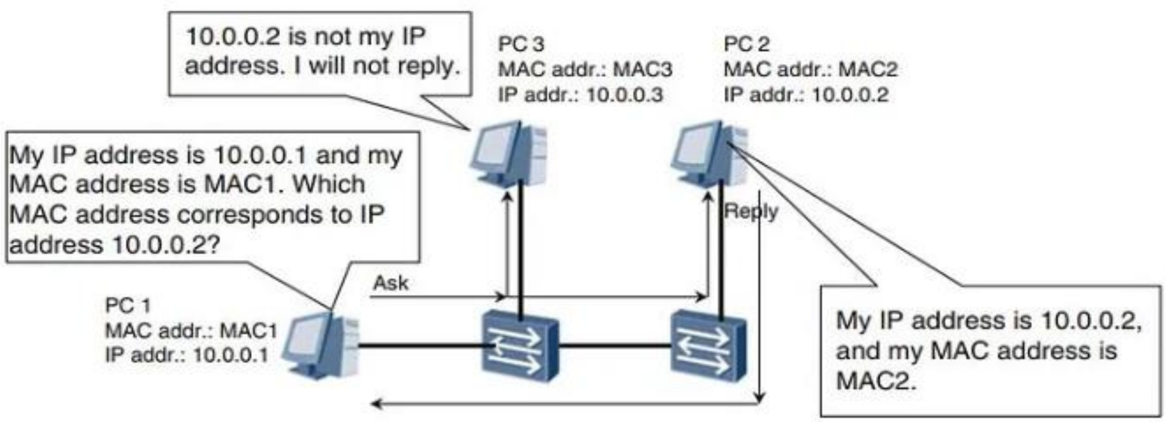
In virtual networking, mapping logical addresses (such as IP addresses) to physical addresses (such as MAC addresses) are typically done through a process called Address Resolution Protocol (ARP). Here's how it works:

Address Resolution Protocol (ARP): ARP is a protocol used to map logical addresses (IP addresses) to physical addresses (MAC addresses) within a local network. When a device needs to send data to another device on the same network, it needs to know the physical address of the recipient.

ARP Request: When a device wants to communicate with another device on the network and knows its IP address but not its MAC address, it sends out an ARP request broadcast packet. This packet contains the IP address of the destination device.

ARP Reply: The device with the corresponding IP address listed in the ARP request responds with an ARP reply packet. This packet contains its own MAC address.

Caching ARP Entries: Once a device receives an ARP reply, it caches the mapping between the IP address and MAC address for future reference. This caching helps reduce ARP traffic on the network and improves performance.



ARP is required to map a logical (IP) address to a MAC address because data link layer devices use MAC addresses to deliver frames within a local network.

The ARP protocol maps a logical IP address to a physical MAC address so that data can be delivered to the correct device in a local network, since switches forward frames using MAC addresses.

In a virtual networking environment, the process of mapping logical addresses to physical addresses through ARP is typically handled by the virtual networking software or hypervisor. Virtual switches and routers within the virtual network act similarly to their physical counterparts in traditional networks, forwarding ARP requests and replies between virtual machines and other network devices.

Here's a brief overview of how mapping logical addresses to physical addresses works in a virtual networking environment:

Virtual Switch: Virtual machines connected to the same virtual switch can communicate with each other using their logical addresses (IP addresses). When a virtual machine needs to send data to another virtual machine on the same virtual switch, it sends an ARP request to resolve the MAC address of the destination virtual machine.

Hypervisor Handling: The hypervisor or virtual networking software intercepts the ARP request and forwards it to the appropriate destination virtual machine. If the destination virtual machine is on the same virtual switch, the hypervisor responds with the MAC address of the destination virtual machine.

Address Caching: Just like in physical networks, virtual machines cache ARP entries to reduce ARP traffic and improve performance.

Virtual Router: If the destination virtual machine is on a different virtual network segment, the ARP request is forwarded to the virtual router or gateway, which performs ARP resolution for devices outside the local network segment.

Overall, the process of mapping logical addresses to physical addresses in virtual networking environments closely resembles the process used in physical networks, with virtual switches, routers, and hypervisors handling ARP requests and replies to facilitate communication between virtual machines.

Lab work (CO2)

1. Execute and comprehend fundamental networking commands within the Ubuntu virtual box environment. Describe how it contributes to diagnosing network connectivity problems, obtaining network-related data, or configuring network settings. Document your observations and explanations in a record

   a) IPConfig
   b) NSLOOKUP
   c) Ping
   d) Tracert
   e) Wget